Topic: Cybersecurity

Topic Title: Cybersecurity in Mobile Banking: Analyzing Phishing Attacks and Security Challenges in Bangladesh

Introduction

The rapid shift toward digital payments and mobile banking has transformed how people manage money globally (Boar & Szemere, 2020). In Bangladesh, this digital growth drives financial inclusion and strongly supports the country's "Smart Bangladesh" vision (Sarker & Khan, 2024). However, this rapid digital transformation has also significantly increased cybersecurity risks for both users and financial institutions (Waliullah et al., 2025). Phishing attacks, malware, and financial fraud frequently target commercial banks because they hold massive amounts of sensitive customer data and capital (Hossain et al., 2025).

To ensure the sustainable growth of digital banking, understanding these security challenges is now essential. This literature review analyzes the current state of mobile banking security by evaluating eight key studies. The review is organized into three major themes: the systemic vulnerabilities of digital banking in Bangladesh, the critical role of human behavior in phishing attacks, and the advanced technical frameworks proposed to mitigate these security risks.

Literature Review

Systemic Vulnerabilities in Bangladesh's Digital Banking Sector

Several researchers highlight that the fast growth of digital banking in Bangladesh has outpaced its security infrastructure. According to Hossain et al. (2025), commercial banks in Bangladesh are primary targets for cybercriminals because many institutions still rely on legacy infrastructure and invest very little in modern cybersecurity. This evaluation is shared by Sijan et al. (2022), who point out that major historical incidents, such as ATM fraud and

the Bangladesh Bank cyber heist, occurred due to outdated systems and a lack of proper automation.

Furthermore, Waliullah et al. (2025) note that while mobile financial services (MFS) and online transfers have boosted the economy, they face constant risks from ransomware and data breaches due to weak authentication mechanisms. On a broader scale, Sarker and Khan (2024) explain that as Bangladesh moves toward a fully digital economy, weak public policy and a lack of skilled professionals make the digital ecosystem highly vulnerable.

Synthesis & Evaluation: Together, these studies show a clear trend: the main issue in Bangladesh is not a lack of digital adoption, but rather a gap between technological expansion and protective infrastructure. While Boar and Szemere (2020) state that global digital payment systems face general infrastructure challenges like network reliability, the situation in Bangladesh is more critical because the foundational security tools are outdated.

The Human Element and the Risk of Phishing Attacks

A major consensus across the literature is that the human factor is the weakest link in mobile banking security. Bojjagani and Sastry (2017) present a threat model showing that security risks do not just exist on bank servers, but are heavily driven by user behavior. Bad habits—such as using weak passwords, clicking suspicious text links, and jailbreaking mobile devices—directly invite phishing and social engineering attacks.

This problem is highly evident in Bangladesh. Sijan et al. (2022) identify customer unawareness as a top threat to local e-banking. Similarly, Sarker and Khan (2024) mention that low public awareness makes everyday users easy targets for identity theft and financial fraud.

Synthesis & Evaluation: These findings demonstrate that even if a bank installs the strongest firewall, a system remains insecure if the user blindly shares their credentials. The studies by Bojjagani and Sastry (2017) and Waliullah et al. (2025) successfully prove that protecting mobile banking is a two-way responsibility. Therefore, focusing only on software updates without educating the public creates an incomplete defense system.

Advanced Technical Solutions and Security Frameworks

To combat these rising threats, researchers have proposed various advanced technological solutions. Admass et al. (2023) discuss how modern technologies like Artificial Intelligence (AI), Machine Learning (ML), and blockchain are currently reshaping global cyber defense by improving real-time threat detection. In Bangladesh, Hossain et al. (2025) suggest that banks must deploy Security Information and Event Management (SIEM) systems and establish dedicated Security Operation Centers (SOC).

Looking at architectural defenses, Sijan et al. (2022) propose a practical three-layer security framework for Bangladeshi banks, separating protection into process, communication, and server-end levels using multi-factor authentication (MFA) and secure VLANs. For highly targeted mobile wallets, Homoliak et al. (2020) introduce a specialized framework called "SmartOTPs," which provides an air-gapped two-factor authentication system using low-cost microcontrollers.

Synthesis & Evaluation: While AI and machine learning offer smart ways to detect modern attacks (Admass et al., 2023), implementing them in Bangladesh requires significant financial investment and technical skill, which current legacy systems lack. However, localized solutions like the three-layer framework by Sijan et al. (2022) or the budget-friendly, offline authentication prototype by Homoliak et al. (2020) offer highly realistic, production-ready defenses against phishing and credential theft that Bangladeshi commercial banks can adopt immediately.

Conclusion

In conclusion, the literature shows that cybersecurity in mobile banking is an ongoing battle between rapid digital growth and evolving cybercrimes like phishing. The studies confirm that the biggest challenges in Bangladesh are outdated banking infrastructure, a shortage of local cybersecurity experts, and low user awareness. While advanced technologies like AI, blockchain, and air-gapped authentication frameworks offer strong protection, they cannot

succeed alone. Future security strategies must combine technical system upgrades with aggressive public awareness programs. Governments, financial institutions, and tech developers must collaborate to build a secure, dependable, and resilient mobile banking environment for the people of Bangladesh.